



Cross-Chain Content Rights Management Service

A Polkadot-Based Unified Rights Token for Subscription, Pay-Per-View, and Permanent Ownership

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A dissertation submitted in partial fulfilment of the requirements for the degree of M.Sc. in Blockchain and Distributed Ledger Technologies (ICT).



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Thank you all.

Abstract

I would first introduce a more generic RQ, that is without the polkadot specific terms, then introduce why polkadot is a suitable platform, then the polkadot specific RQ

The content creator economy concentrates value in centralized intermediaries; no existing system delivers subscription, pay-per-view, and permanent-ownership access rights through a single cross-chain-portable primitive. This dissertation asks: *how can a Polkadot parachain built on FRAME pallet primitives and XCM deliver a unified rights token natively supporting all three monetization models across heterogeneous blockchain networks, and what levels of transaction finality, cost efficiency, and creator revenue retention can such a system achieve?*

CCRMS (Cross-Chain Content Rights Management Service) is a Polkadot parachain implementing parent-child NFT nesting on pallet-nfts, automatic royalty splits, and cross-chain portability via XCM and Snowbridge. The prototype is evaluated with 56 unit tests, seven performance benchmarks, a security audit, a centralized Express.js/SQLite baseline comparison, and a 10,000-iteration Monte Carlo creator revenue simulation.

The abstract is meant to give the reader an idea, this is a bit too specific, very quickly you should contextualise the work for the reader

All 11 KPIs met their targets: 26.2 TPS sustained throughput, ~6-second inclusion latency, 3–5-block cross-chain finality, 191 bytes per content item, 90% baseline uptime, and 15–20-second MTTR. The Monte Carlo simulation yields ~16.8% higher average creator revenue than centralized platforms, but this advantage is **strongly**

audience-dependent: creators above ~10,000 followers benefit decisively, while those below ~2,000 are better served by centralized discovery platforms.

The abstract should read more as an abstract for the reader that has not read your work, then give exact detail where required

Three findings advance the state of knowledge. First, a unified rights token supporting all three access models can be constructed from standard FRAME primitives without novel cryptographic mechanisms, closing literature gap G7. Second, the revenue advantage of decentralized rights management is not universal, crossing a follower threshold at approximately 2,000. Third, the work demonstrates that conservatively chosen architectural decisions—specifically the pallet-first design—provide incidental resilience against ecosystem-level changes, as evidenced by the January 2026 discontinuation of ink! development.

Keywords: cross-chain content rights, Polkadot XCM, unified rights token, creator revenue retention, design science research

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List of Abbreviations

ABI	Application Binary Interface
AI	Artificial Intelligence
API	Application Programming Interface
ASCII	American Standard Code for Information Interchange
BLS	Boneh–Lynn–Shacham (signature scheme)
CAGR	Compound Annual Growth Rate
CCRMS	Cross-Chain Content Rights Management Service
CLI	Command-Line Interface
CPU	Central Processing Unit
DEX	Decentralized Exchange
DOT	Polkadot native token
DRM	Digital Rights Management
DSR	Design Science Research
ESMA	European Securities and Markets Authority
ETH	Ether (Ethereum’s native token)
EU	European Union
EVM	Ethereum Virtual Machine
FRAME	Framework for Runtime Aggregation of Modularized Entities
GB	Gigabyte
H160	160-bit hash (Ethereum-style address)
HHI	Herfindahl-Hirschman Index
HRMP	Horizontal Relay-routed Message Passing
HTTP	Hypertext Transfer Protocol
ID	Identifier

IPFS	InterPlanetary File System
JAM	Join-Accumulate Machine
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
MEV	Maximal Extractable Value
MiCA	Markets in Crypto-Assets (EU Regulation)
MTTR	Mean Time To Recovery
NFT	Non-Fungible Token
OTT	Over-The-Top (media service)
PIL	Programmable IP License (Story Protocol)
PolkaVM	Polkadot Virtual Machine
PPV	Pay-Per-View
RISC	Reduced Instruction Set Computer
RMRK	Remark (NFT protocol; derived from the <code>system.remark</code> extrinsic)
RPC	Remote Procedure Call
SCALE	Simple Concatenated Aggregate Little-Endian (Substrate codec)
SDK	Software Development Kit
SIGTERM	Signal Terminate (POSIX signal)
SQL	Structured Query Language
TLS	Transport Layer Security
TPS	Transactions Per Second
TVL	Total Value Locked
UI	User Interface
USD	United States Dollar
WASM	WebAssembly
XCM	Cross-Consensus Messaging
ZK	Zero-Knowledge (proof / cryptography)

Introduction

1.1 | The Problem

The prevailing content economy is unfavorable to creators. An independent musician earning USD 100,000 in streaming revenue typically retains between USD 55,000 and USD 70,000 after platform fees and distributor commissions — a figure that is substantially lower for label-signed artists (Music Business Research, 2024). Centralized intermediaries retain between 30 and 60 percent of gross creator revenue, depending on the platform and contract structure; the per-platform breakdown used in our analysis is presented in Section 7.2.3 (Table 7.12). Fees are non-negotiable, audiences are non-portable, and content rights remain bound by the platform's terms of service throughout.

Consumers face similar fragmentation. A film bought on iTunes cannot be streamed on Amazon Prime; an Audible audiobook cannot be transferred to Google Play; a subscription started on one platform cannot be moved to another. Consumers do not own their purchases; they hold a platform-bound license that can be revoked, modified, or ignored. The fragmentation that constrains creators also restricts the consumers who support them.

Fragmentation extends to monetization models. Subscriptions typically live on Patreon; one-time sales on Gumroad; pay-per-view is scarcely supported because the overhead of conventional payment systems makes micro-transactions uneconomic. Each platform supports a single monetization model, has its own audience, charges its own fees, and produces its own accounting reports. A creator who wants to offer all three models is forced to run three or four platforms in parallel and reconcile them by hand.

Existing blockchain based content distribution platforms?

Existing **blockchain alternatives** address parts of the problem but not the whole.

Non-fungible tokens (NFTs) provide transferable ownership but not recurring access or per-view consumption. Subscription-style decentralized platforms (such as Audius)

You'll need to introduce what has been proposed, both from an industry perspective, but more importantly from a technical/academic perspective so

that your work can be contextualised within them.

This should be done in background/literature review

offer recurring revenue but are confined to a single chain and audience. Cross-chain bridges move assets between ecosystems but lack metadata for rights. Blockchain-based digital rights management has been examined in the academic literature specifically for the music industry (Ciriello et al., 2023), and proposed solutions share a common limitation: they address rights metadata or royalty distribution in isolation, without combining all three monetization models into a single portable primitive. No existing system, centralized or decentralized, delivers subscription, pay-per-view, and permanent ownership from a single rights primitive that is portable across blockchains.

The technical primitives to address this (cross-chain messaging, programmable smart contracts, on-chain royalty distribution, and NFT-based rights tokenization) already exist as standalone capabilities. Hanneke et al. (2024) characterize this class of systems as an emerging *Internet of Value*, in which blockchain tokenization enables asset ownership and revenue flows that bypass centralized intermediaries entirely. **Our research examines whether these primitives can be integrated into a coherent system that benefits creators more than existing solutions and the conditions under which creators would adopt it.**

You must demonstrate//convince examiners that therefore, this has not yet been done; and also why the approach you propose is a suitable way to do so

1.2 | Background Context

This dissertation sits at the intersection of three research and engineering areas: blockchain-based digital rights management, cross-chain interoperability, and decentralized content monetization. Each area is surveyed in Chapter 2; this section provides the context needed to motivate the work.

You already provided the motivation for the industries in the intro... I would avoid doing more of this

The global DRM market stands at approximately USD 6.7 billion in 2025; the blockchain-specific segment is growing at 33% CAGR, projected to reach USD 2.74 billion by 2030 (Chapter 2 surveys both).

Now focus on diving into what has been proposed; where they are lacking (in intro/background/lit review) then later in your chapters you need to convince examiners that what you propose is academically novel; why the decisions made are sound; and how you answer the RQ

Cross-chain interoperability has emerged as an active research area during the implementation window. A 2023 survey in IEEE Transactions on Knowledge and Data Engineering (Ren et al., 2023) classifies interoperability approaches into five categories — sidechains, notary schemes, hashed time lock contracts, relays, and blockchain-agnostic protocols — and two further 2025 surveys (Deng et al., 2025; Li et al., 2025b) collectively cover more than 150 sources. The Polkadot ecosystem that underpins our technical foundation completed its 2.0 upgrade during this period, materially improving per-parachain latency and throughput; the Snowbridge V2 and JAM developments are surveyed in Chapter 2.

Why is this important to mention here? Will the examiners even understand what ink is ?

In January 2026, the ink! Alliance discontinued development of ink!, the language selected for the CCRMS contract layer. Our pallet-first architecture proved resilient to the

OK, but again that doesn't mean much to examiners at this point/

Again, not sure of relevance here.

change (Appendix A.9). The EU AI Act copyright provisions (August 2025) and MiCA (full enforcement July 2026) create regulatory demand for cryptographic provenance and tamper-evident rights records, capabilities that CCRMS can provide.

A Master's research academic dissertation is not just about proposing, designing/implementing but demonstrating that what is being proposed is novel (academically)

1.3 | Research Aim and Approach

OK, but first need to convince the reader that this hasn't been done, and why this is a good solution to the problem

Our objective is to **design, implement, and evaluate a functional prototype of a cross-chain content rights management service that supports three monetization models: subscription, pay-per-view, and permanent ownership, using a single unified rights primitive with automatic royalty distribution and cross-chain portability**. We designate this prototype CCRMS (the Cross-Chain Content Rights Management Service) and implement it as a Polkadot parachain.

What do you mean?

The primary research question that guides our work is:

I would introduce this, first then discuss design/implementation/evaluation etc

How can a shared-security, multi-chain framework built on Polkadot's Cross-Consensus Messaging (XCM) and FRAME pallet primitives deliver a unified rights token that natively supports recurring subscriptions, pay-per-view microtransactions, and permanent ownership transfers across heterogeneous blockchain networks, and what levels of transaction finality, cost efficiency, and creator revenue retention can such a system achieve?

I still feel you need to contextualise the problem, and convince the reader that this has not been done, yet and why this configuration/proposal is suitable to help (just at an abstract level)

The question has two mutually constraining components: design feasibility and measurement — which are fully decomposed in Chapter 3.

Our research approach is grounded in **design science research** (Hevner et al., 2004) and applied to a multi-component blockchain prototype. We organize the methodology, detailed in Chapter 4, into iterative build-test-measure-refine cycles within a six-phase framework. Our evaluation strategy, as delineated in Chapter 6, combines quantitative measures, including technical benchmarks, security audits, comparative analyses, and Monte Carlo simulations of creator revenue, with qualitative insights from literature synthesis. We frame the evaluation as **measurement-focused** rather than target-driven: our objective is to document and analyze the levels of performance, security, and economic viability achieved by the system, rather than to verify conformity with a predefined set of criteria.

1.4 | Contributions

We make four contributions, described in detail in Section 9.1:

1. **A feasibility result for the unified rights primitive.** This dissertation demonstrates that a single FRAME pallet, with a unified priority-resolution access check across three storage maps, is sufficient to encode subscription, pay-per-view, and permanent ownership as native access modes for a single token type — without novel cryptographic or consensus-layer mechanisms. This closes the gap G7 identified in Chapter 2.
2. **Quantified decentralization–latency and revenue trade-offs.** The latency cost of decentralization is measured directly: CCRMS incurs a $67,000\times$ overhead against a centralized baseline. The creator-revenue advantage is real but conditional: CCRMS produces higher mean revenue only for creators with approximately 2,000 followers or more, the point at which algorithmic discovery no longer compensates for higher platform fees.
3. **A cross-chain rights artifact demonstrating XCM sufficiency.** The CCRMS parachain — comprising two FRAME pallets, an ink! 6 wrapper contract, an EVM precompile, and benchmark and simulation scripts — demonstrates that XCM v3 with payer–beneficiary decoupling is sufficient for authenticated cross-chain rights operations without custom protocol extensions. The artifact is released under the MIT license.
4. **A DSR worked example for multi-component blockchain prototypes.** Applying design science research to a system spanning FRAME pallets, smart contracts, XCM, and Monte Carlo simulation is rarely illustrated in the existing DSR literature; this dissertation serves as a reference template for future researchers.

1.5 | Scope and Limitations

This dissertation is bound in several ways that the reader should be aware of before engaging with the main content chapters.

The prototype is a back-end service, not a consumer application. We did not develop a web or mobile interface for CCRMS. Our assessment focuses on pallet- and runtime-level behavior; we do not directly evaluate user-experience factors. We recognize this as a limitation in Chapters 6 and 8 and as an area for future research in Chapter 9.

We take measurements on a local development network. Our CCRMS prototype runs in a local Zombienet topology with two relay-chain validators and single-collator parachains. In production, parachains are hosted on the Polkadot or Kusama relay chains, which have hundreds of validators, real-world network latency, and competition from

many other parachains. The throughput, latency, and reliability metrics in Chapter 7 should be treated as **upper bounds for the test environment**, not forecasts of real-world deployment performance.

Our Snowbridge integration uses a local Ethereum stack. We successfully demonstrated the bridge with two transactions, each transferring one ETH, from a local Ethereum chain to a local AssetHub parachain. However, the entire configuration remains local; no public testnet or mainnet has been used. This integration serves as a proof of architectural feasibility rather than a production validation.

We have not commissioned any third-party audit, security review, or independent benchmarking. A single author performed the implementation, test design, measurements, and interpretation for this dissertation. Deployment to production would require third-party validation before any substantive value is processed.

